

Abstraction

Abstraction is the process of hiding the implementation details and showing only the necessary functionality to the user.

The user knows what an object does, but not how it does it.

Real-Time Example - 1

Consider an ATM Machine.

You only know:

- Insert Card
- Enter PIN
- Withdraw Money
- Check Balance

You don't know:

- How PIN is verified
- How bank server communicates
- How money is counted
- How database is updated

The internal implementation is hidden. This is called Abstraction.

Real-Time Example – 2

Consider a Fan.

You only know:

- Turn ON the Fan
- Turn OFF the Fan
- Increase Speed
- Decrease Speed

You don't know:

- How the motor rotates
- How electricity is converted into mechanical motion
- How the speed regulator controls the motor
- How the internal electrical components work

The internal implementation is hidden. This is called Abstraction.

Why Abstraction?

Abstraction is used to

- Hide complex implementation
 - Improve security
 - Reduce code complexity
 - Improve maintainability
 - Achieve standardization
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How Python Achieves Abstraction

Python does not support abstraction directly.

In order to perform abstraction we need to import ABC and abstractmethod from the abc module.

ABC

- It stands for Abstract Base Class.
 - It is a special base class (also called a helper class) provided by Python in the abc (Abstract Base Classes) module.
 - It is used to create abstract classes and enforce abstraction.
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abstractmethod

- Is a function provided by Python's abc module.
- It is used as a decorator to declare abstract methods inside an abstract class.

Python provides abstraction using:

1. Abstract Class
2. Abstract Methods

Conditions to Make a Class Abstract

A class becomes an abstract class if

- It inherits from ABC
- It contains at least one abstract method

What is a Abstract Class?

An Abstract Class is a class that contains one or more abstract methods. It acts as a blueprint for child classes.

What is a Concrete Class?

A Concrete Class is a class that implements all inherited abstract methods and whose objects can be created.

What is a Abstract Method?

An Abstract Method is a method declared inside an abstract class decorated with @abstractmethod which has no implementation .

What is a Concrete Method (Normal Method)?

A Concrete Method (also called a Normal Method) is a method that has a complete implementation (method body) and performs a specific task without decoration.

What Can an Abstract Class Contain?

- Class Variables
- Instance Variables
- Constructor
- Abstract Methods (mandatory ,atleast one)
- Concrete Methods (normal methods usually normal instance method)
- class and static methods

Important Notes

- Concrete methods are optional in abstract class.
- Abstract methods are mandatory in abstract class.
- Multiple abstract and concrete methods are allowed .
- Child class must implement all abstract method.
- Abstract Class cannot be created with a Object , object should be created for child class.

Why Can't We Create an Object for an Abstract Class?

Because abstract methods have no implementation. Python does not know what those methods should do. Therefore object creation is restricted.

Child Class Rules

If parent class has 3 abstract methods then child class must implement all 3 methods with same names.

Real-Time Examples

- ATM Machine
- Car Driving
- Mobile Phone

- Television Remote
- Washing Machine
- Payment Gateway
- Online Banking
- UPI Applications
- Printer
- Coffee Machine

Difference Between Abstract Class and Concrete Class

Abstract Class	Concrete Class
Cannot create object	Object can be created
Contains abstract method(s)	No abstract methods
Used as blueprint	Used for implementation
Inherits from ABC	May or may not inherit

Difference Between Abstract Method and Concrete Method

Abstract Method	Concrete Method
No implementation (typically uses pass)	Has implementation
Decorated with @abstractmethod	No decorator required
Must be overridden by child class	Overriding is optional
Declares what should be done	Defines how it is done

Interview Questions

1. What is abstraction and Why do we use abstraction?
 2. How is abstraction achieved in Python?
 3. What is abc module?
 4. What is ABC and @abstractmethod?
 5. Can we create an object of an abstract class?
 6. Can an abstract class have static methods and class methods?
 7. What happens if a child class doesn't implement all abstract methods?
 8. What is the difference between abstraction and encapsulation?
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